

Circuit Theory Foundations

Sections 1–16 • cross-referenced to EE 105 Lectures 5–8

Problems are grouped by topic. Full solutions to the problems in Parts A–G, in matching order, begin on page 14; Problems 24–34 (Parts H–I, Sections 15–16) are practice problems without printed solutions. Problem 35 (Part I, page 13) has a full class-note solution on page 18, including the supporting concepts. Unless stated otherwise, treat all resistors as ideal and all sources as ideal.

A. Charge, Current, Voltage, and Power

Problem 1 conceptual

Explain the difference between conventional current and electron flow. If a formula gives a current of $+2\text{ A}$ to the right, which way are the electrons actually drifting?

Problem 2 analytical

A device carries a steady 0.40 A for 5 minutes. (a) How much charge passes? (b) How many electrons is that?

Problem 3 analytical

A 9 V battery delivers 150 mA to a circuit. (a) What power does it supply? (b) How much energy (in joules) does it deliver in 10 minutes?

Problem 4 conceptual

Voltage is defined “between two points,” yet we routinely say “the voltage at node A is 3 V .” What unstated convention makes that statement meaningful?

B. Resistance and Ohm's Law

Problem 5 conceptual

State the difference between the *definition* $R = V/I$ and *Ohm's law* $V = IR$. For which kind of element does only the first one apply?

Problem 6 analytical

A resistor drops 3.3 V when 15 mA flows through it. (a) Find R . (b) Find the power it dissipates. (c) Would a $\frac{1}{8}$ W resistor survive?

Problem 7 analytical

Two wires are the same material and length, but wire B has twice the cross-sectional area of wire A. What is the ratio R_B/R_A ? If wire A is instead stretched to twice its length (area unchanged), what is its new resistance relative to the original?

Problem 8 Python

Write a short script that plots I versus V for a $470\ \Omega$ resistor over $V \in [0, 5]$ V, and on the same axes plots the (nonohmic) relation $I = I_s(e^{V/V_T} - 1)$ for a diode with $I_s = 10^{-12}$ A, $V_T = 25.85$ mV. Comment on why only one of the two is a straight line.

C. Series, Parallel, and the Dividers

Problem 9 analytical

Find the equivalent resistance of $1\text{ k}\Omega$, $2\text{ k}\Omega$, and $2\text{ k}\Omega$ all in parallel.

Problem 10 analytical

A voltage divider has $V_{\text{in}} = 12\text{ V}$, $R_1 = 8.2\text{ k}\Omega$ (top) and $R_2 = 3.3\text{ k}\Omega$ (bottom). (a) Find V_{out} (across R_2). (b) Now connect a $3.3\text{ k}\Omega$ load across the output. Recompute V_{out} and comment on the change.

Problem 11 analytical

A 10 mA source feeds two parallel branches, $R_1 = 1\text{ k}\Omega$ and $R_2 = 4\text{ k}\Omega$. Use the current-divider rule to find the current in each branch. Which branch carries more, and why?

Problem 12 conceptual

Explain in one or two sentences why the parallel combination of two resistors is always smaller than either one.

D. Sources, Measurement, and Faults

Problem 13 conceptual

A real 1.5 V battery has internal resistance $r_s = 0.5\text{ }\Omega$. Sketch (in words) why its terminal voltage drops as the load draws more current, and compute the terminal voltage into a $2.5\text{ }\Omega$ load.

Problem 14 conceptual

Why is an ammeter placed in series and a voltmeter in parallel? What input resistance would each ideally have, and what goes wrong if a real meter departs from the ideal?

Problem 15 conceptual

Distinguish a short circuit from an open circuit in terms of resistance and resulting current, and explain how a fuse uses one to prevent damage from the other.

E. Kirchhoff's Laws and Nodal Analysis

Problem 16 analytical

A 10 V source drives three resistors in series: $R_1 = 1\text{ k}\Omega$, $R_2 = 2\text{ k}\Omega$, $R_3 = 2\text{ k}\Omega$. Use KVL to find the loop current and the voltage across each resistor. Verify the drops sum to 10 V.

Problem 17 analytical

A 12 V source connects through $R_1 = 2\text{ k}\Omega$ to node A . From A , $R_2 = 4\text{ k}\Omega$ goes to ground and $R_3 = 4\text{ k}\Omega$ goes to ground. Use nodal analysis to find V_A and the source current.

Problem 18 Python

Set up the nodal equations for a two-node resistive network of your choice as a matrix equation $\mathbf{G}\mathbf{v} = \mathbf{i}$ and solve it with `numpy.linalg.solve`. Print the node voltages and verify KCL numerically at each node.

F. Superposition, Thévenin, and Norton

Problem 19 conceptual

When applying superposition, how do you “turn off” a voltage source? A current source? Why does the method fail if the circuit contains a diode?

Problem 20 analytical

For the divider $V_{\text{in}} = 9\text{ V}$, $R_1 = 4.7\text{ k}\Omega$ (top), $R_2 = 10\text{ k}\Omega$ (bottom), output across R_2 : find the Thévenin equivalent (V_{TH} , R_{TH}) seen at the output terminals, then use it to predict V_{out} when a $10\text{ k}\Omega$ load is attached.

Problem 21 analytical

Convert the Thévenin equivalent from the previous problem into its Norton equivalent (I_{N} , R_{TH}). Confirm $V_{\text{TH}} = I_{\text{N}}R_{\text{TH}}$.

G. Capacitors and Inductors

Problem 22 analytical

A $2.2\ \mu\text{F}$ capacitor charges through a $22\ \text{k}\Omega$ resistor from a step input. (a) Find the time constant. (b) Roughly how long until it is $> 99\%$ charged?

Problem 23 conceptual

State the duality between capacitors and inductors. Which quantity can “not jump” for each element, and which defining equation expresses that?

H. Batteries (Section 15)

Problems 24–34 were previously numbered 15.1–16.9 in the Sections 14–26 problem set; the original number is shown in the tag.

Problem 24 analytical ▪ was 15.1

A battery with a capacity of 1800 mAh is to be used in a device that draws 120 mA continuously. Ignoring possible loss in capacity as a result of load current magnitude, how long should the battery be able to deliver power?

Problem 25 analytical ▪ was 15.2

Determine the discharge time and average current output of a battery with a capacity rating of 1000 mAh if it is discharged at 1 C.

How long would it take to discharge at 5 C, 2 C, 0.5 C, 0.2 C and 0.05 C?

I. Resistor Networks and Circuit Analysis (Section 16)

Problem 26 analytical ■ was 16.1

If a $1000\ \Omega$ resistor is connected in parallel with one of $3000\ \Omega$, what is the total (equivalent) resistance?

Problem 27 analytical ■ was 16.2

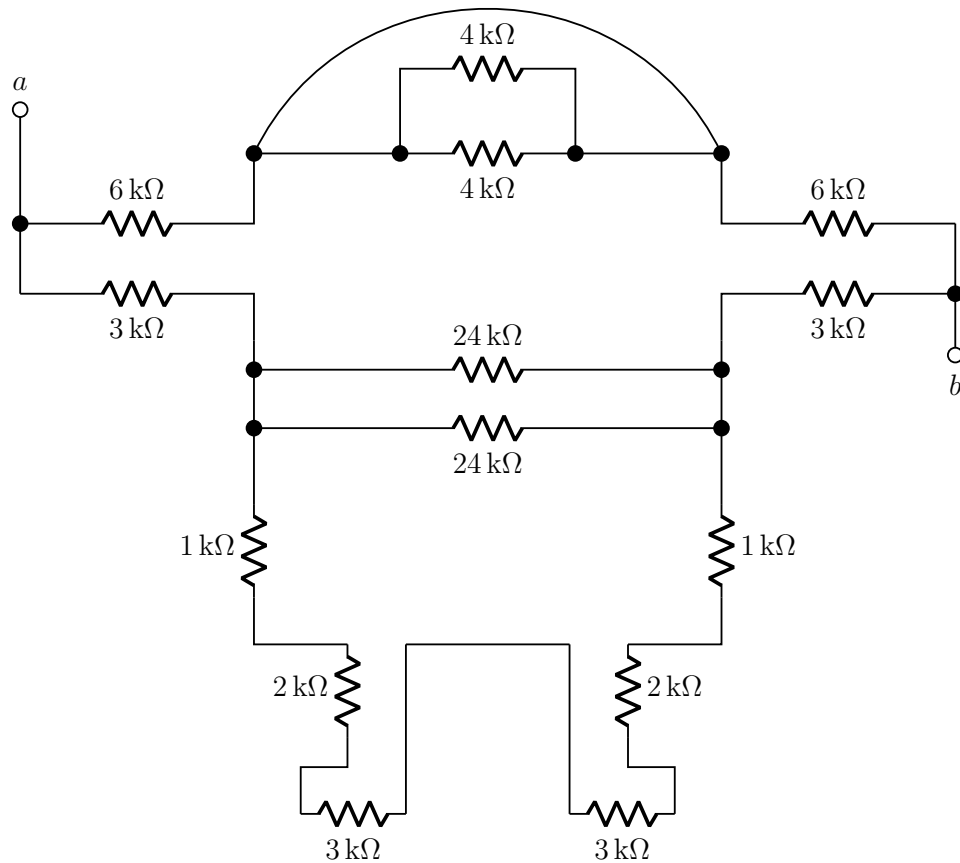
If a $1.0\ \text{k}\Omega$ resistor is placed in series with a $2.0\ \text{k}\Omega$ resistor, what is the total resistance?

If the two resistors are then attached to an input voltage of $V_{in} = 9\ \text{V}$, find:

- (a) The total current flow and the current through each resistor.
- (b) The voltage drop across each resistor.

Problem 28 analytical ■ was 16.3

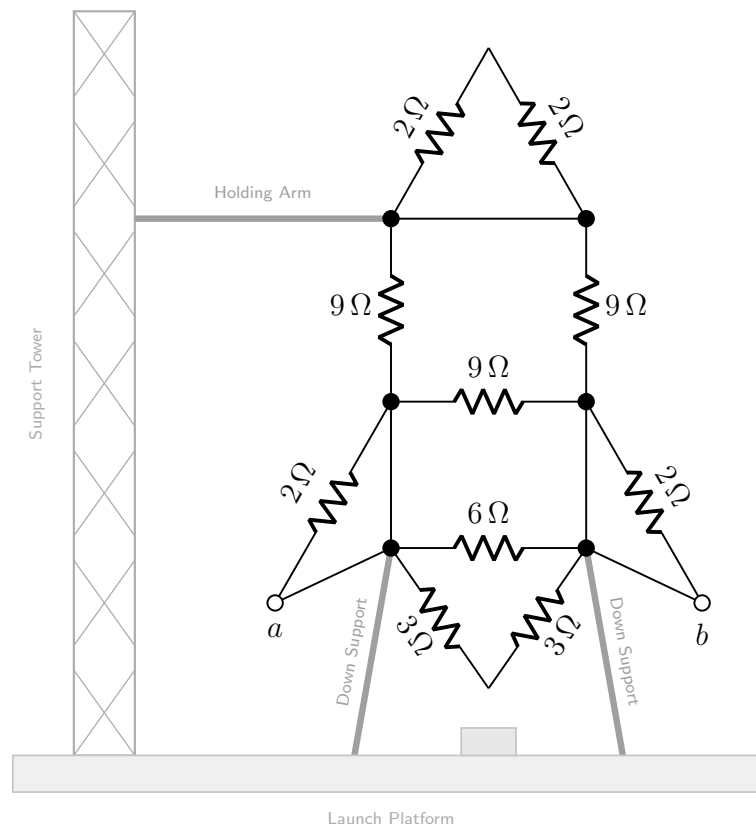
Find the equivalent resistance between terminals a and b of the resistor network shown below.



Problem 29 analytical ■ was 16.4

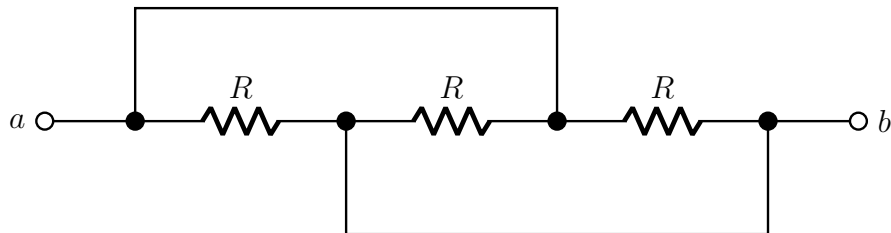
Find the equivalent resistance between terminals a and b of the resistor network shown below. All resistances are given in ohms (Ω).

The support tower, holding arm, down supports and launch platform (drawn in grey) are mechanical structures only — they are **not** electrical connections.

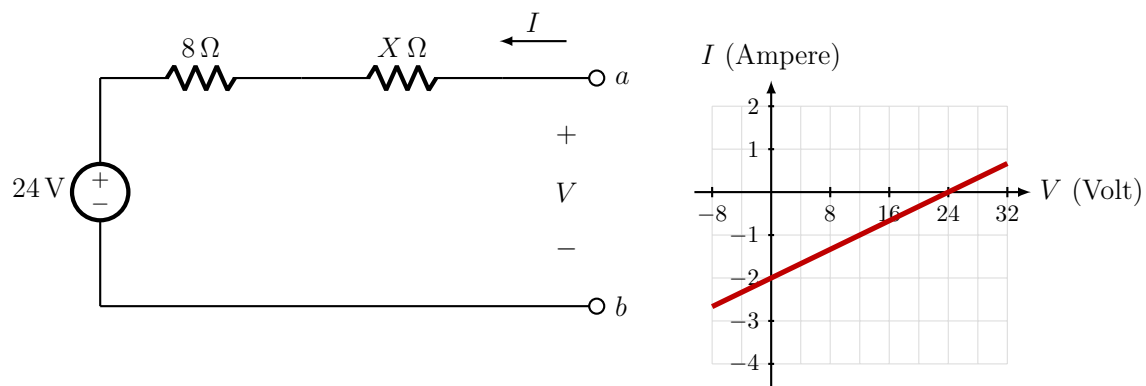


Problem 30 analytical ■ was 16.5

In the circuit below, all three resistors have the same resistance R . Find the equivalent resistance between the terminals a and b . What is the configuration of the three resistors — series, parallel, or neither?

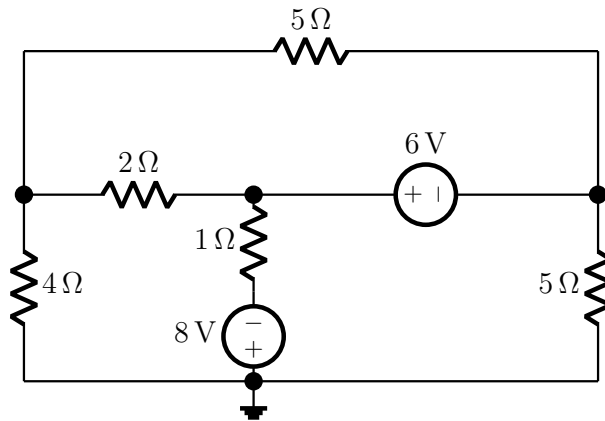
**Problem 31** analytical ■ was 16.6

The two-terminal network below consists of a 24 V source in series with an $8\ \Omega$ resistor and an unknown resistance X . The terminal current I and the terminal voltage V , measured at terminals a – b with the polarities marked in the figure, obey the straight-line relationship shown alongside. Determine the value of the unknown resistance X .



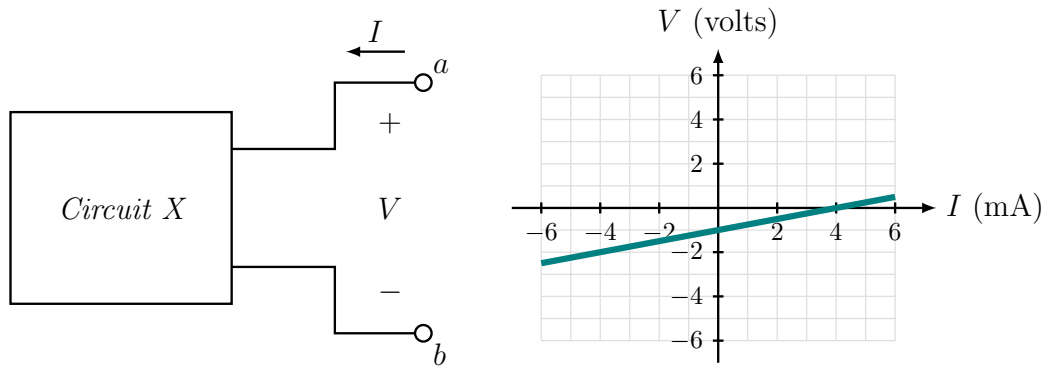
Problem 32 analytical ■ was 16.7

Apply nodal (or mesh) analysis to the circuit below to determine all of the node voltages — or, if you choose to work with meshes, all of the mesh currents. The bottom rail, marked with the ground symbol, is the reference node.

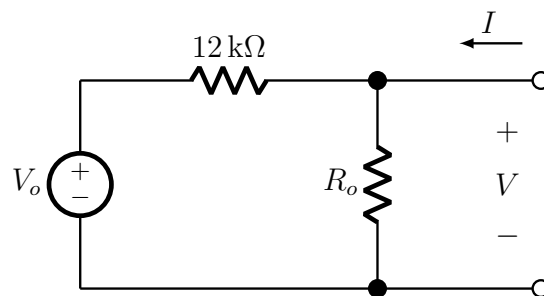


Problem 33 analytical ■ was 16.8

Circuit X is a linear two-terminal circuit connected between terminals a and b , as shown below. When tested, the circuit follows the voltage (V) and current (I) relationship shown in the accompanying plot.

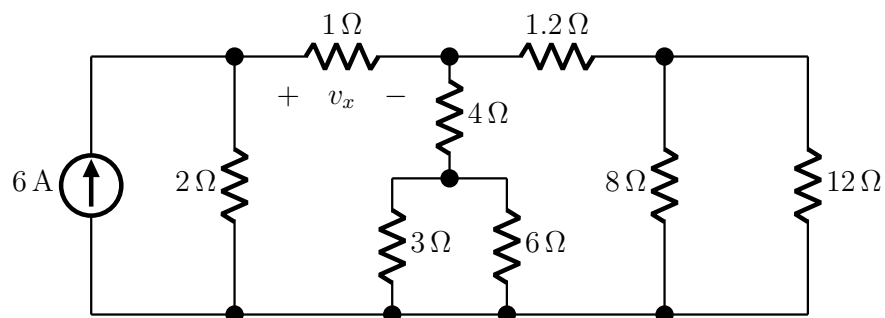


- Identify the range of the voltage V for which the circuit behaves as an **active** circuit (i.e. supplying power).
- Analyse the network below to determine the values of R_o and V_o , assuming this network is equivalent to *Circuit X*.



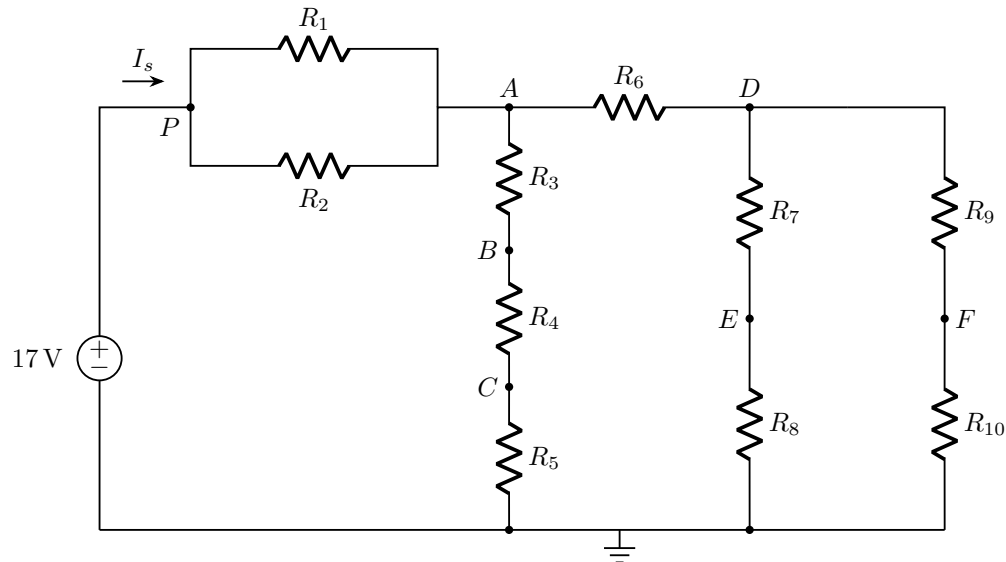
Problem 34 analytical ■ was 16.9

Find the voltage v_x across the $1\ \Omega$ resistor in the circuit below, with the polarity marked (+ on the left). The source supplies a constant 6 A.



Problem 35 analytical ■ class example, September 22, 2026

All ten resistors are $1\text{ k}\Omega$. All sources and wires are ideal. Node voltages are measured relative to ground. Use successive series and parallel reductions.



- Find the equivalent resistance seen by the 17 V source. Show and label each intermediate reduction.
- Find the total current I_s supplied by the source.
- Find the voltages at nodes A, B, C, D, E, and F. Show your calculations.

Solutions

Solution 1

Conventional current is the direction *positive* charge would move; electron flow is opposite. A +2 A current to the right means electrons drift to the *left*.

Solution 2

(a) $Q = It = 0.40 \times 300 = 120 \text{ C}$. (b) $120 / (1.602 \times 10^{-19}) \approx 7.5 \times 10^{20}$ electrons.

Solution 3

(a) $P = VI = 9 \times 0.150 = 1.35 \text{ W}$. (b) $E = Pt = 1.35 \times 600 = 810 \text{ J}$.

Solution 4

A single node voltage is understood *relative to a chosen 0 V reference* (ground). “ $V_A = 3 \text{ V}$ ” means the potential difference between A and ground is 3 V.

Solution 5

$R = V/I$ is the *definition* of resistance and holds for any element point-by-point. $V = IR$ with constant R (Ohm’s law) holds only for *ohmic* elements. For a nonohmic element (diode, filament), only the definition applies.

Solution 6

(a) $R = V/I = 3.3/0.015 = 220 \Omega$. (b) $P = VI = 3.3 \times 0.015 = 49.5 \text{ mW}$. (c) A $\frac{1}{8} \text{ W} = 125 \text{ mW}$ resistor is rated well above 49.5 mW, so yes.

Solution 7

$R \propto L/A$. Doubling area halves resistance: $R_B/R_A = 0.5$. Doubling length (area fixed) doubles resistance; if the wire is stretched so volume is conserved the area would halve and R would rise $\times 4$, but with area held fixed the answer is $\times 2$.

Solution 8

The resistor line is straight because $I = V/R$ is linear (constant slope $1/R$). The diode’s exponential $I(V)$ curves sharply upward, so it is nonohmic — no single resistance describes it.

```
import numpy as np, matplotlib.pyplot as plt
V = np.linspace(0, 5, 400)
I_res = V/470.0
Is, VT = 1e-12, 0.02585
I_diode = Is*(np.exp(V/VT)-1)
```

```
plt.plot(V, I_res*1e3, label='470 ohm (ohmic)')
plt.plot(V, np.clip(I_diode,0,0.05)*1e3, label='diode (nonohmic)')
plt.xlabel('V (V)'); plt.ylabel('I (mA)'); plt.legend(); plt.show()
```

Solution 9

$1/R = 1/1000 + 1/2000 + 1/2000 = 0.002 \text{ S}$, so $R = 500 \Omega$.

Solution 10

(a) $V_{\text{out}} = 12 \cdot 3.3 / (8.2 + 3.3) = 12 \cdot 0.287 = 3.44 \text{ V}$. (b) Load $3.3 \text{ k}\Omega \parallel 3.3 \text{ k}\Omega = 1.65 \text{ k}\Omega$; $V_{\text{out}} = 12 \cdot 1.65 / (8.2 + 1.65) = 12 \cdot 0.168 = 2.01 \text{ V}$. The output sags because the divider is not a stiff source — loading it lowers the bottom-leg resistance.

Solution 11

$I_1 = I R_2 / (R_1 + R_2) = 10 \cdot 4 / (1 + 4) = 8 \text{ mA}$; $I_2 = 10 \cdot 1 / 5 = 2 \text{ mA}$. The *smaller* resistor (R_1) carries the *larger* current — current prefers the easier path.

Solution 12

Adding a parallel path gives the current more ways to flow, which can only increase total conductance; larger conductance means smaller resistance than either branch alone.

Solution 13

Terminal voltage $= V_S R_L / (R_L + r_s) = 1.5 \cdot 2.5 / (2.5 + 0.5) = 1.25 \text{ V}$. As R_L falls, more current flows, so more voltage is dropped across r_s , leaving less at the terminals.

Solution 14

An ammeter must carry the branch current, so it goes in series and ideally has 0Ω (no added drop). A voltmeter must see the potential difference, so it goes in parallel and ideally has infinite resistance (no stolen current). A real meter's non-ideal resistance loads the circuit and biases the reading, worst when comparable to the circuit resistance.

Solution 15

A short is $R \rightarrow 0$, giving very large current ($I = V/R$); an open is $R \rightarrow \infty$, giving zero current. A fuse carries normal current but melts into an *open* when current gets too large (e.g. from a short), interrupting the circuit before damage spreads.

Solution 16

$I = 10/(1000 + 2000 + 2000) = 2 \text{ mA}$. Drops: $IR_1 = 2 \text{ V}$, $IR_2 = 4 \text{ V}$, $IR_3 = 4 \text{ V}$; sum = 10 V. ✓

Solution 17

KCL at A: $(V_A - 12)/2\text{k} + V_A/4\text{k} + V_A/4\text{k} = 0$. $\times 4\text{k}$: $2(V_A - 12) + V_A + V_A = 0 \Rightarrow 4V_A = 24 \Rightarrow V_A = 6 \text{ V}$. Source current = $(12 - 6)/2\text{k} = 3 \text{ mA}$.

Solution 18

Example network (node voltages v_1, v_2):

```
import numpy as np
# Conductance matrix G v = i (siemens, amps)
G = np.array([[1/1000+1/2000, -1/2000],
              [-1/2000, 1/2000+1/4000]])
i = np.array([12/1000, 0.0]) # 12 V source via 1k into node 1
v = np.linalg.solve(G, i)
print("node voltages:", v)
```

Substituting v back into each row of $\mathbf{Gv} - \mathbf{i}$ should give ≈ 0 , confirming KCL.

Solution 19

Turn off a voltage source by replacing it with a *short* (0 V); turn off a current source by replacing it with an *open* (0 A). Superposition needs linearity ($V = IR$ with constant R); a diode is nonlinear, so responses do not add and the method fails.

Solution 20

$V_{\text{TH}} = \text{open-circuit voltage} = 9 \cdot 10 / (4.7 + 10) = 6.12 \text{ V}$. $R_{\text{TH}} = R_1 \parallel R_2 = (4.7 \cdot 10) / (4.7 + 10) = 3.20 \text{ k}\Omega$ (source shorted). With a $10 \text{ k}\Omega$ load: $V_{\text{out}} = 6.12 \cdot 10 / (10 + 3.20) = 4.64 \text{ V}$.

Solution 21

$I_{\text{N}} = V_{\text{TH}}/R_{\text{TH}} = 6.12/3200 = 1.91 \text{ mA}$, with the same $R_{\text{TH}} = 3.20 \text{ k}\Omega$ in parallel. Check: $I_{\text{N}}R_{\text{TH}} = 1.91 \text{ mA} \times 3.20 \text{ k}\Omega = 6.12 \text{ V} = V_{\text{TH}}$. ✓

Solution 22

(a) $\tau = RC = 22\,000 \times 2.2 \times 10^{-6} = 48.4 \text{ ms}$. (b) $> 99\%$ charged after about $5\tau \approx 242 \text{ ms}$.

Solution 23

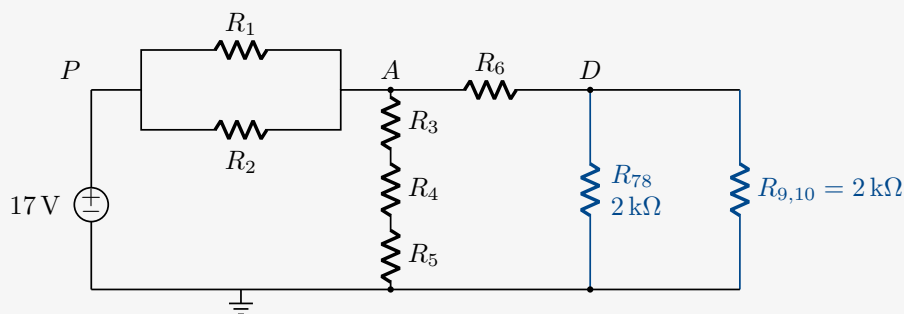
Capacitor and inductor are duals ($V \leftrightarrow I$, $C \leftrightarrow L$). A capacitor's *voltage* cannot jump (from $I = C dV/dt$); an inductor's *current* cannot jump (from $V = L dI/dt$).

Solution 35**Part A — Equivalent resistance**

Blue components show the reduction made in each step. All unreduced resistors are $1\text{ k}\Omega$; node 0 is ground.

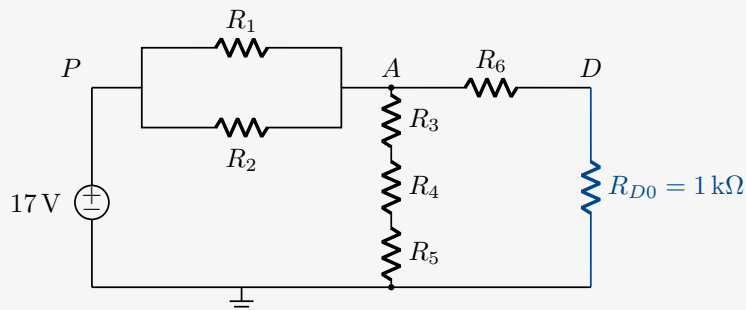
Step 1. Combine the two right-hand series pairs.

$$R_{78} = R_7 + R_8 = 2\text{ k}\Omega, \quad R_{9,10} = R_9 + R_{10} = 2\text{ k}\Omega.$$



Step 2. Combine the D-to-ground branches in parallel.

$$R_{D0} = R_{78} \parallel R_{9,10} = 2\text{ k}\Omega \parallel 2\text{ k}\Omega = 1\text{ k}\Omega.$$

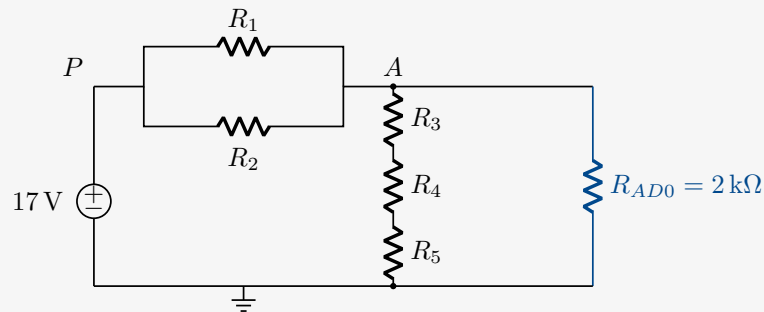


Connection test: series elements carry the same current; parallel branches connect to the same two nodes and have the same voltage.

Part A — Continue reducing

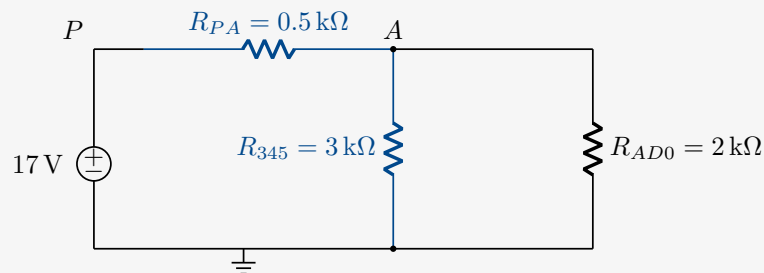
Step 3. Combine R_6 with the reduced D-to-ground network.

$$R_{AD0} = R_6 + R_{D0} = 1\text{ k}\Omega + 1\text{ k}\Omega = 2\text{ k}\Omega.$$



Step 4. Reduce the central series chain and the input parallel pair.

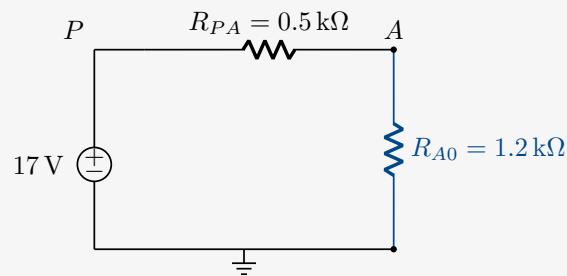
$$R_{345} = R_3 + R_4 + R_5 = 3\text{ k}\Omega, \quad R_{PA} = R_1 \parallel R_2 = 0.5\text{ k}\Omega.$$



Part A — Final two reductions

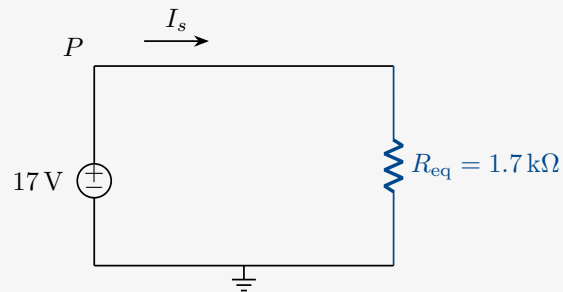
Step 5. Combine the two A-to-ground branches.

$$R_{A0} = 3\text{ k}\Omega \parallel 2\text{ k}\Omega = \frac{3 \times 2}{3 + 2}\text{ k}\Omega = 1.2\text{ k}\Omega.$$



Step 6. Combine the remaining series resistances.

$$R_{\text{eq}} = R_{PA} + R_{A0} = 0.5\text{ k}\Omega + 1.2\text{ k}\Omega = 1.7\text{ k}\Omega.$$



Part B — Total source current

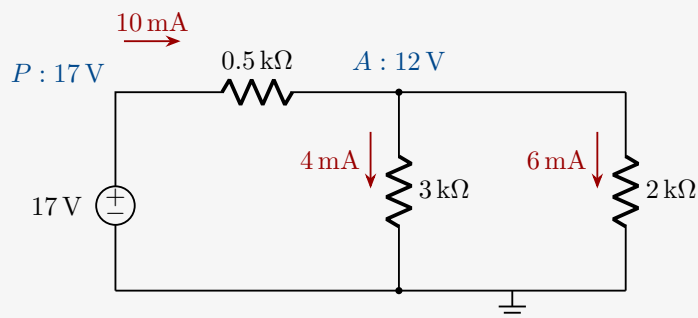
$$I_s = \frac{17\text{ V}}{1.7\text{ k}\Omega} = 10\text{ mA}.$$

Units: $\text{V}/\text{k}\Omega = \text{mA}$, and $\text{mA} \cdot \text{k}\Omega = \text{V}$.

Part C — First find the branch currents

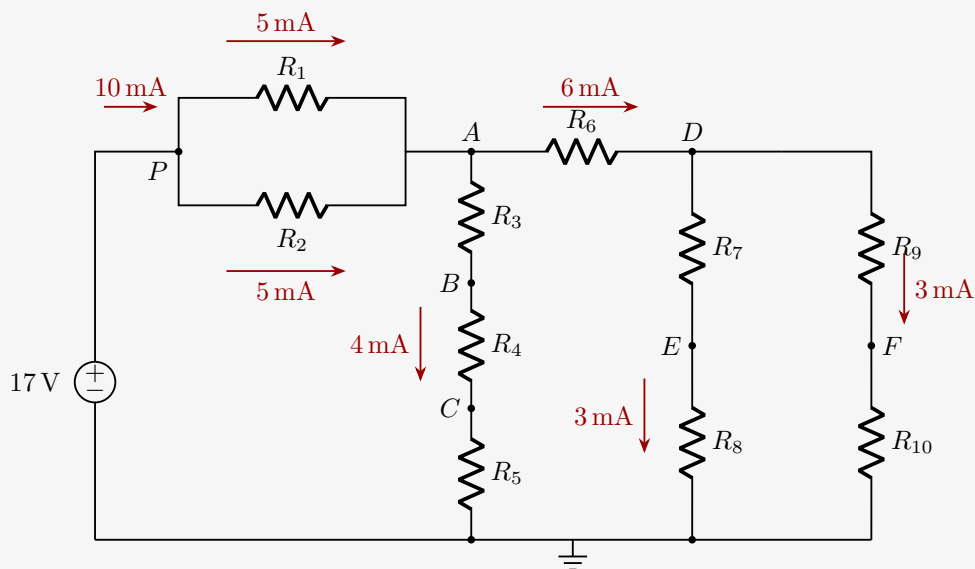
Restore the Part A, Step 4 circuit. The drop from P to A is $I_s R_{PA} = 5\text{ V}$:

$$V_A = 17\text{ V} - (10\text{ mA})(0.5\text{ k}\Omega) = 12\text{ V}.$$



$$I_{3k} = 10 \frac{2}{3+2} = 4\text{ mA}, \quad I_{2k} = 10 \frac{3}{3+2} = 6\text{ mA}.$$

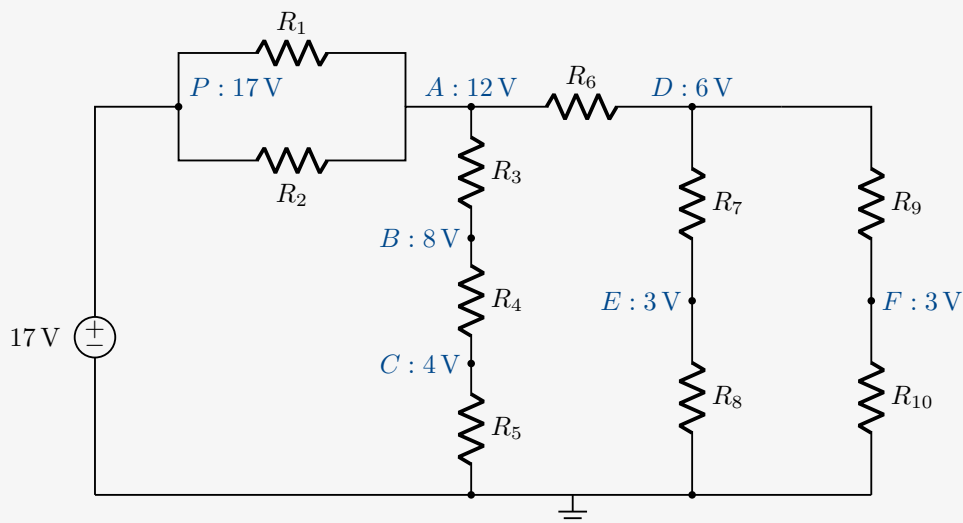
Restore the original circuit. Red arrows give the current directions and values.



At P, 10 mA splits equally through R_1 , R_2 . At D, 6 mA splits equally into the two $2\text{ k}\Omega$ branches. Every resistor in an unbranched series path carries the same current.

Part C — Then recover the node voltages

Move along each current arrow and subtract IR . Blue labels show voltages relative to ground.



$$V_A = 17 - (10)(0.5) = \boxed{12\text{ V}},$$

$$V_B = 12 - (4)(1) = \boxed{8\text{ V}},$$

$$V_C = 8 - (4)(1) = \boxed{4\text{ V}},$$

$$V_D = 12 - (6)(1) = \boxed{6\text{ V}},$$

$$V_E = 6 - (3)(1) = \boxed{3\text{ V}},$$

$$V_F = 6 - (3)(1) = \boxed{3\text{ V}}.$$

The products above use mA and $\text{k}\Omega$, giving volts. KCL checks: $10 = 4 + 6$ at A and $6 = 3 + 3$ at D (mA).

Compact reference

Series: same current

$$R_{\text{eq}} = \sum R_j; n \text{ equal resistors: } nR.$$

Parallel: same two nodes

$$1/R_{\text{eq}} = \sum 1/R_j; n \text{ equal resistors: } R/n.$$

Voltage division (series)

$$V_k = V_{\text{in}} R_k / \sum R_j.$$

Current division (parallel)

$$I_k = I_{\text{in}} (1/R_k) / \sum (1/R_j) = I_{\text{in}} G_k / \sum G_j.$$

Two parallel branches

$$I_a = I_{\text{in}} R_b / (R_a + R_b); G = 1/R \text{ (conductance)}.$$

Trace wires, not shapes: every point on an ideal wire is one node. These four resistors are all parallel between α and β :

